Will Donovan



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Academic positions

2021–	Yau MSC, Tsinghua University, Beijing: Associate Professor
2018–21	Yau MSC, Tsinghua University, Beijing: Assistant Professor
2014–18	Kavli IPMU, University of Tokyo: Postdoctoral Fellow
2011–14	University of Edinburgh: Research Assistant

Further affiliation

2021–	Yanqi Lake BIMSA: Adjunct Associate Professor
2020–	Kavli IPMU, University of Tokyo: Visiting Scientist

Education

2007–11	PhD Mathematics, Imperial College London (with R.P. Thomas, E. Segal)
2003–4	MMath Mathematics, University of Cambridge (Part III)
2000–3	BA Mathematics, University of Cambridge

Publications

Papers

1. Grassmannian twists on the derived category via spherical functors
   Proc. London Math. Soc. (2013) 107 (5): 1053–1090.

2. Window shifts, flop equivalences and Grassmannian twists (with E. Segal)
   Compositio Math. (2014) 150 (6): 942–978.

3. Mixed braid group actions from deformations of surface singularities (with E. Segal)
   Comm. Math. Phys. (2015) 335 (1): 497–543.

4. The Pfaffian-Grassmannian equivalence revisited (with N. Addington and E. Segal)
   Alg. Geom., Foundation Compositio Math. (2015) 2 (3): 332–364.

5. Noncommutative deformations and flops (with M. Wemyss)
   Duke Math. Jour. (2016) 165 (8): 1397–1474.

6. Moduli spaces of torsion sheaves on K3 surfaces (with N. Addington and C. Meachan)
   J. London Math. Soc. (2016) 93 (3): 846–865.

7. Mukai flops and $\mathbb{P}$-twists (with N. Addington and C. Meachan)
   J. Reine Angew. Math. (Crelle) (2019) 748: 227–240.

8. Contractions and deformations (with M. Wemyss)
   Amer. J. Math. (2019) 141 (3): 563–592.

9. Twists and braids for general 3-fold flops (with M. Wemyss)
   J. Eur. Math. Soc. (2019) 21 (6): 1641–1701.

10. Perverse schobers and wall crossing
    Int. Math. Res. Notices (2019) 18: 5777–5810.

11. Noncommutative enhancements of contractions (with M. Wemyss)
    Adv. Math. (2019) 344: 99–136.

12. Perverse schobers on Riemann surfaces: constructions and examples
    Eur. J. Math. (2019) 5: 771–797.

13. Mirror symmetry for perverse schobers from birational geometry (with T. Kuwagaki 桑垣樹)
    Comm. Math. Phys. (2021) 381: 453–490.

14. Stringy Kähler moduli for the Pfaffian-Grassmannian correspondence
    SIGMA (2021) 17, 028.

15. Relating derived equivalences for simplices of higher-dimensional flops
    Adv. Stud. Pure Math. (2023) 88: 305–330.

16. Root stacks and periodic decompositions (with A. Bodzenta)
    Manuscripta Math. (2024) 175: 53–73.

17. Stringy Kähler moduli, mutation and monodromy (with M. Wemyss)
    J. Diff. Geom. (2025) 129 (1): 115–164.

18. Derived symmetries for crepant contractions to hypersurfaces
    Preprint, `arXiv:2409.19555`.

19. Window categories for a simple 9-fold flop of Grassmannian type
    (with W. Hara, M. Kapustka, and M. Rampazzo)
    Preprint, `arXiv:2510.06184`.

20. Quiver braid group action for a 3-fold crepant resolution (with Luyu Zheng 郑璐予)
    Preprint, `arXiv:2512.19140`.

21. D-equivalence conjecture for hyperkähler varieties of generalized Kummer type
    (with Chengxi Wang 王成茜 and Nantao Zhang 张南涛)
    Preprint, `arXiv:2611.04118`.

22. Higher-dimensional conifolds and hypergeometric systems (with Weilin Su 苏蔚琳)
    In preparation.

23. Categorical crepant contractions and spherical functors
    In preparation.

Proceedings

24. Grassmannian twists, derived equivalences and brane transport
Proc. Symp. Pure Math. (2015), Proc. String-Math 2012.

25. Contractions of 3-folds: deformations and invariants
Internat. J. Math. (2016) 27 (7), special issue VBAC 2014.

26. Applications of noncommutative deformations
Proc. Kinosaki Alg. Geom. Symp. (2016), available online.

Grants

2019– China central government **Thousand Talents** Plan
Value: 2.5 million yuan (∼$350k)

2016–18 Japan Society for Promotion of Science KAKENHI **Young Scientist** B
Value: 2.9 million yen (∼$25k)

Awards

2020 Yau MSC Ruolin 若琳 **Research Excellence** Award

2019 Yau MSC **Outstanding Paper** Award 'Twists and braids for general 3-fold flops'

Postgraduate students

Doctoral

2021–26 **Nantao Zhang** 张南涛 birational and enumerative geometry, Bridgeland stability
'Stability conditions on local Calabi–Yau fourfolds and on blowups'
Postdoctoral researcher, Sun Yat-Sen University (2026–)

2020–25 **Luyu Zheng** 郑璐予 toric geometry, braid groups, perverse sheaves
'Braid groups associated with exceptional surfaces in 3-folds'
Postdoctoral researcher, SCMS, Fudan University (2026–)

2019–23 **Xun Lin** 林汛 derived categories, birational geometry, Hodge theory
'Non-commutative aspects of Hodge conjecture, minimality, and Torelli problems'
Assistant professor, CUHK Shenzhen (2026–)
Postdoctoral fellow, HKUST & MPIM Bonn (2023–26)

Masters

2019–25 **Weilin Su** 苏蔚琳 categorification, hypergeometric systems, mirror symmetry
'Derived monodromy for the Appell F_4 hypergeometric system'

Examining

2024 **Caroline Namanya** doctoral external examiner
'Pure braids and group actions from 3-fold flops'
Makerere University, Uganda / Glasgow University, UK

Mentoring

Junior faculty

2023–25 **Beihui Yuan** 袁北彗
Assistant professor, BIMSA, Beijing (2023–)

Postdocs

2026– **Marco Rampazzo**

2024–26 **Alicia Lamarche**
Assistant professor, BIMSA, Beijing (2026–)

2024–26 **Aimeric Malter**
Assistant professor, HIMIS, Shenzhen (2026–)

Teaching

Yau MSC, Tsinghua University

2026	Topics in derived categories
2026	Theoretical mechanics
2025	Topics in complex algebraic geometry
2025	Topics in algebraic geometry (homological mirror symmetry)
2024	Theoretical mechanics
2023	Topics in algebraic geometry (derived categories and toric geometry)
2023	Calculus 2
2022	Theoretical mechanics
2022	Lie groups and Lie algebras
2021	Geometric representation theory 2 (with Penghui Li 李鹏辉)
2021	Derived functors in algebraic and birational geometry
2020	Calculus 1
2020	Mirror symmetry: categories and constructions
2019	Geometric representation theory 2 (with Peng Shan 单芃)
2019	Calculus 2
2018	Derived functors in algebraic and birational geometry

Yanqi Lake BIMSA

2023	Introduction to topos quantum theory
2022	Quiver approaches to machine learning
2022	Categorical and geometric aspects of quantum mechanics

Mini Courses

2027	Makerere U, Uganda	'Crepant contractions and spherical functors'
2025	Yonsei U, Seoul	'Derived symmetries, crepant contractions and noncommutative deformations'
2020	KIAS, Seoul	'Perverse schobers and mirror symmetry'
2012	Steklov Inst, Moscow	'Symmetry and the derived category'

Extended visits

Apr–May 2024	Sydney Mathematical Research Institute
Mar 2024	IMS, Chinese University of Hong Kong
Jun–Jul 2020	Korean Institute of Advanced Study, Seoul
Feb–Jun 2020	Kavli IPMU, University of Tokyo
Jan–Apr 2014	Hausdorff Institute, Bonn
Apr–Jul 2013	Schrödinger Institute, Vienna
May 2012	Steklov Institute, Moscow

Service

Organization

Summer school 'Perverse sheaves and their categorification in geometry and representation theory' at Venice International University, organizer (2026)

Workshop 'Homological algebra of the infrared' at TSIMF, Sanya, organizer (2023)
∼20 speakers, including A. Bondal, M. Kapranov, Fields medallist M. Kontsevich

Workshop 'Derived categories' at Hausdorff Institute, Bonn, organizer (2014)

Yau MSC, Tsinghua

Institution Representative, China-Africa Mathematics Center (2026–)
Foreign affairs committee (2023–)
Tutor, Qiuzhen college (2023–)
Events Hub organizer (2021–)
Geometric representation theory seminar organizer (2021–)
Colloquium committee (2019–21)

Outreach

– Funckup night 'Always be calculating', Beijing (2026)

– Workshop series, for University of Tokyo 'Soap bubbles and spacetime' (2017–)
At 6 locations, including Tamarokuto Science Centre and Kavli IPMU

– Interviewed, for University of Tokyo 'Discover Our People' (2016)

– Talk: 'The frontiers of symmetry: hidden geometry and tangled superstrings'
King's College London Math School (2015)
Summerhall, Edinburgh, for International Year of Crystallography (2014)

Previous employment

2006–7	Lead programmer, Bloomberg LP: financial research and development
2005	TEFL teacher, Kyoto, Japan

Languages

– Japanese (JLPT N2), Mandarin (advanced)
– English (native)

Selected talks

2026
ICBS, Beijing
IMS, CUHK, Hong Kong

2025
Cetraro, Italy
Wuhan U, China

2024
Fujiyoshida, Japan
Makerere U, Uganda
UBC, Vancouver
TSIMF Sanya, Hainan

2023
IMPAN, Warsaw
Kavli IPMU, Tokyo
Krakow, Poland
Max Planck Inst, Bonn

2022
Cambridge U
Oberwolfach
UCL, London

2021
BICMR, Peking U
Cardiff U
CMSA, Harvard U

2020
IASM, Hangzhou
KAIST, Korea
Steklov Inst, Moscow

2019
Auckland U, New Zealand

2019 (continued)
Australian National U
BIRS Oaxaca, Mexico
CTU, Prague
Fudan U, Shanghai
Kyoto U
Liverpool U
Osaka U
Sydney U
SYSU, Guangzhou

2018
Hokkaido U
Nagoya U
TSIMF Sanya, Hainan

2017
Australian National U
Bath U
ICMS, Edinburgh U
Nagoya U
Osaka U
SISSA, Trieste

2016
BIRS Banff
ECM, Berlin
Freie Universität, Berlin
Hanga Roa, Chile
Hong Kong U
Kinosaki Symposium
Kyoto U
METU, Ankara

2015
Nagoya U
Fields Institute, Toronto
Max Planck Inst, Bonn
Oberwolfach
Warwick U

2014
Freie Universität, Berlin
BICMR, Beijing
Hamburg U
KIAS, Seoul
Sheffield U

2013
Cardiff U
Columbia U
Duke U
Edinburgh U
Imperial College London
Leeds U
Oxford U
Schrödinger Inst, Vienna

2012
ETH Zürich
Glasgow U
Hausdorff Institute, Bonn
Newcastle U
Queen Mary, U London
Warwick U